

Axiolyn Lead-Hunting Automation

Project brief and implementation plan — n8n workflow (self-hosted, zero-cost stack)

1. Project overview

Axiolyn (axiolyn.com) is a business run by Waleed and friends, offering three service lines: Business Management (process and SOP design, KPI dashboards, workflow architecture), Software Solutions (custom web and internal apps, CRM integrations, APIs and data pipelines), and AI Automation (chatbots, agents, workflow automation, document and data processing). Right now client acquisition is manual. This project builds an automated lead-hunting pipeline in n8n that continuously finds companies, in Pakistan or internationally, that are likely to need one of these services, and appends their details to a Google Sheet for the team to contact.

2. Goal

Build an n8n automation that, on a schedule, searches for companies matching Axiolyn's target profile, extracts their contact information (company name, website, email, phone, location, industry, source, date found), filters out duplicates already in the sheet, and appends only new leads to a shared Google Sheet, with no manual searching required.

3. Constraints (must follow)

- n8n is self-hosted locally — no n8n Cloud subscription. The workflow must run on a local, self-hosted n8n instance.
- Zero budget — no paid APIs or subscriptions of any kind (no Apollo.io, no Hunter.io paid tier, no SerpApi paid plan, etc). Every component must run on a genuinely free tier or be built from free, open tools.
- Google Sheets is the storage layer, accessed via the user's own free Google account (OAuth2 credentials in n8n), not a paid Sheets or Workspace add-on.
- The pipeline should be resumable and idempotent — re-running it should never create duplicate rows for a company already logged.

4. Proposed architecture

Six-stage n8n workflow, all nodes either built into n8n or backed by free-tier services:

Stage	n8n node(s)	Free approach
1. Trigger	Schedule Trigger	Runs daily or a few times a week. Built into n8n, no cost.

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2. Search for leads	HTTP Request	Google Custom Search JSON API (100 free queries/day, no billing setup needed) targeting industry and region queries, and/or scraping public business directories (Pakistan Yellow Pages, Clutch, GoodFirms) directly by URL.
3. Extract listings	HTML Extract (Cheerio, built in)	Parse company names, website URLs, and addresses out of the raw HTML. No external service.
4. Filter by fit	IF / Filter node	Keep only results matching target industries, size, and region; drop clearly irrelevant results.
5. Enrich contacts	HTTP Request + Code node (regex)	Fetch each company's homepage or contact page and regex-extract email and phone patterns directly from the page text. Replaces paid email-finder APIs.
6. Dedup + append	Google Sheets (Lookup, Append Row)	Look up the company or domain in the existing sheet; only append if not already present. Fully free with a personal Google account.

5. Google Sheet schema

One row per company, one sheet acting as the lead database. Suggested columns:

- Company name
- Website URL
- Email (regex-extracted)
- Phone (regex-extracted)
- Location / country
- Likely service fit (Business Management / Software Solutions / AI Automation)
- Source (which search query or directory it came from)
- Date found
- Contact status (empty by default, filled in manually as the team follows up)

6. Search strategy and target profile

Companies worth targeting typically show signs like: no visible CRM or automation, manual-looking intake processes, an outdated website, active hiring for roles that automation would reduce, or industries known for operational sprawl. Good starting query angles:

- Industry + region combos, e.g. "logistics company" Lahore contact, "clinic" Karachi, "real estate agency" site:.pk
- Directory scraping for agencies and SMBs already listed as looking for digital transformation (Clutch, GoodFirms public listing pages)

- Industries Axiolyn already lists as a fit: e-commerce and retail, healthcare and clinics, logistics and supply chain, real estate, professional services, agencies and consultancies, education and training, SaaS and tech startups

7. Suggested implementation phases

- Phase 1 — Foundation: set up the local n8n instance, create the Google Sheet with the schema above, connect Google Sheets OAuth2 credentials in n8n.
- Phase 2 — Search and extraction: build the Schedule Trigger to Google Custom Search (or directory scrape) to HTML Extract chain. Verify it returns clean company name and website pairs.
- Phase 3 — Enrichment: add the per-company HTTP Request and regex Code node to pull email/phone from each site's contact page. Handle sites that fail to load or have no visible contact info gracefully, without crashing the workflow.
- Phase 4 — Dedup and write: add the Google Sheets lookup step and the conditional Append Row step so re-runs never duplicate a company.
- Phase 5 — Test and tune: run manually a few times, check lead quality and false positives, tune filter criteria and search queries.
- Phase 6 — Schedule it: turn on the Schedule Trigger for ongoing daily or weekly runs.

8. Notes and open considerations

Google Custom Search's free tier caps at 100 queries a day, so this produces a steady trickle of leads rather than mass scraping — reasonable for a small team's actual outreach capacity. Regex-scraped emails and phone numbers will be somewhat noisier than a paid verification API, so a quick manual skim of new rows before outreach is expected. Keep outreach to public, business-facing contact info rather than scraping personal emails, and be mindful of each source site's terms of service when scraping directories.